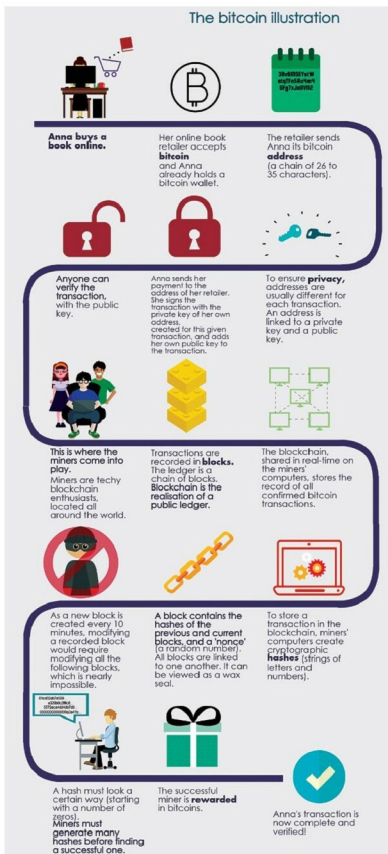




transactions. The payment system enables users to pay each other directly, without having to rely on a third party. Since Bitcoin is a digital cryptocurrency, it cannot be contained physically. There had to be an effective way to manage all transactions in the system, without giving control to any single party. This is where blockchain comes in. It is a distributed public ledger that records all transactions in a particular system," says Abhishek.

ToneTag follows a similar approach to Bitcoin blockchain, but the application varies. The way in which payment transactions are recorded at present is inefficient. The current system involves many entities and players, which leaves greater room for fraud. Every year hundreds of billions of dollars are lost due to fraudulent transactions. To address these security issues, ToneTag introduced blockchain technology to contactless payments. Blockchain technology has the potential to completely redefine the way transactions take place, as it brings unprecedented advancements in fraud prevention.

"ToneTag payments are tokenised, where a unique identifier replaces sensitive transaction information and none of the actual customer or merchant transaction information is revealed or shared, hence making the entire process highly-secure and fraud-proof," says Abhishek.

ToneTag enables contactless payments that are traceable, transparent and secure. Every party involved in the transaction is protected by the blockchain since it maintains a public record of every transaction that ever took place in the system. While paying through ToneTag, customers can identify whether the merchant outlet they are dealing with is exposed to risks or if it has witnessed fraudulent transactions in the past. Similarly, merchants can ensure whether a customer is genuine or not. Further, all transactions are validated to ensure these are authorised.

Miners, the quiet workers

We keep saying that, in a distributed ledger, data is stored in a distributed, decentralised way, spanning multiple digital locations across the globe. What makes this possible? The answer is mining.

Fig. 2: Illustration showing how a blockchain works (Image courtesy: European Payment Council)